

Quantum Optimal Control via Semi-Automatic Differentiation

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We develop a framework of semi-automatic differentiation that enables efficient gradient-based methods of quantum optimal control with automatic differentiation. The approach allows to optimize periodically non-interacting Hamiltonian and is implemented in the open-source Julia package `QOOC` [1] and Vantage [2] (part of the `QuantumOptim.jl` [3] ecosystem). Our method is based on formally treating the optimal control pulse as a time-dependent parameter, enabling efficient control of quantum gates, or quantum gates, in the context of applications of the theory to their design to optimize the time propagation of the evolution of the Hamiltonian when solving the equation. The formalism is extended to the case of a system with a fixed number of particles. The latter is achieved by introducing a continuous parameter, α , into the system with a physically motivated constraint on the parameter. This approach demonstrates the potential accuracy and efficiency of semi-automatic differentiation and facilitates efficient gradient descent in quantum control by enabling the direct optimization of experimental observables for quantum initialization and quantum readout, typically non-interfering observables. We show the full reachability of the set of equilibrium differential states for the optimization of periodic interacting quantum gates in superconducting qubit coupled to external readout lines. This includes the full direct optimization of the Hamiltonian of the system.

1 Introduction

Quantum control was central to the development of quantum technologies [4–7]. Quantum information processing [8], quantum simulation [9], and quantum control [10] all rely on the ability to manipulate matter at the fundamental quantum level. In engineering, analytical solutions to the control problem may be available [11]. In more realistic settings, in particular when taking into account classical or quantum control [12] or quantum feedback [13], the problem must be solved. The most general methods for quantum feedback control, gradient-based optimization [14]–[16] and Krotov’s method [17–19], naturally handle the full range of control possible via arbitrary waveform generation [20] or digital pulse shaping [21]. These methods iteratively optimize the control on partial information, that is, the existence of an optimization functional with respect to the control pulse shape. More precisely, the control parameters are the amplitudes of time-dependent control fields, e.g., laser pulses in the control of trapped atoms in ions, and microwave pulses for the control of superconducting circuits.

Traditionally, the optimal problem has to be defined analytically. In particular the objective, the numerical scheme [22], is formulated specifically for an overlap with a target state. This latter setting was implemented above [23–25] in a small class of optimization frameworks that include stochastic gradient and gradient-free. The implementation of Krotov’s method [26] provides efficient